MATH 152 **QUIZ 10** NAME Solutions

Mrs. Bonny Tighe 25 points

11.10 – 11.11 TA_____Mon 11/19/2018

Due at the beginning of discussion on Monday 11/26/2018

1. Evaluate the limit using series $\lim_{x\to 0}\frac{\tan^{-1}(2x)-2x}{2x-\sin(2x)}$

$$\tan^{-1}(2x)=\sum_{n=0}^{\infty}\frac{(-1)^n(2x)^{2n+1}}{2n+1}=2x-\frac{2^3x^3}{3}+\frac{2^5x^5}{5}-\frac{2^7x^7}{7}\cdots$$

$$\sin(2x)=\sum_{n=0}^{\infty}\frac{(-1)^n(2x)^{2n+1}}{(2n+1)!}=2x-\frac{2^3x^3}{3!}+\frac{2^5x^5}{5!}-\frac{2^7x^7}{7!}$$

$$\lim_{x\to 0}\frac{2x-\frac{2^3x^3}{3}+\frac{2^5x^5}{5}-\frac{2^7x^7}{7}-\cancel{2x}}{2x-\left(2x-\frac{2^3x^3}{3!}+\frac{2^5x^5}{5!}-\frac{2^7x^7}{7!}\right)}=$$

$$\lim_{x\to 0}\frac{-\frac{2^3x^3}{3}+\frac{2^5x^5}{5}-\frac{2^7x^7}{7}\cdots}{\frac{2^3x^3}{3!}-\frac{2^5x^5}{5!}+\frac{2^7x^7}{7!}\cdots}=\lim_{x\to 0}\frac{-\frac{2^3}{3}+\frac{2^5x^2}{5}-\frac{2^7x^4}{7}\cdots}{\frac{2^3}{3!}+\frac{2^5x^2}{5!}+\frac{2^7x^4}{7!}\cdots}$$

$$\frac{-2^3/3}{2^3/3!}=-\frac{1}{3}\cdot\frac{6}{1}=-2$$

2. Use the Binomial Theorem to expansion of the function as a power series.

$$f(x)=\sqrt[3]{8+2x^4}=\left(8\left(1+\frac{x^4}{4}\right)\right)^{1/3}=8^{1/3}\left(1+\frac{x^4}{4}\right)^{1/3}=2\left(1+\frac{x^4}{4}\right)^{1/3}$$

$$\sum_{n=0}^{\infty}2\square\left(\frac{x^4}{4}\right)^n$$

$$\sum_{n=0}^{\infty}2\square\frac{x^{4n}}{2^{2n}}$$

$$\square=1+k+k(k-1)+k(k-1)(k-2)\cdots$$

$$=1+\tfrac{1}{3}+\tfrac{1}{3}\left(-\tfrac{2}{3}\right)+\tfrac{1}{3}\left(-\tfrac{2}{3}\right)\left(-\tfrac{5}{3}\right)+\tfrac{1}{3}\left(-\tfrac{2}{3}\right)\left(-\tfrac{5}{3}\right)\left(-\tfrac{8}{3}\right)$$

$$=\underset{n=0}{1}+\underset{n=1}{\frac{1}{3}}-\underset{n=2}{\frac{2}{3^2}}+\underset{n=3}{\frac{2\cdot 5}{3^3}}-\underset{n=4}{\frac{2\cdot 5\cdot 8}{3^4}}\cdots$$

$$\frac{(-1)^{n+1}(2\cdot 5\cdot 8\cdots(3n-4))}{3^n}$$

$$\boxed{2+\frac{2}{3}\frac{x^4}{2^2}+\sum_{n=2}^{\infty}\frac{(-1)^{n+1}(2\cdot 5\cdot 8\cdots(3n-4))}{3^n\,2^{2n-1}}x^{4n}}$$

3. Find the MacLaurin series using a known series

a) $f(x) = 4\cos(2x^2)$

$$f(x) = 4\sum_{n=0}^{\infty} \frac{(-1)^n (2x^2)^{2n}}{(2n)!}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n+2} x^{4n}}{(2n)!}$$

b) $f(x) = x^2 e^{4x}$

$$f(x) = \sum_{n=0}^{\infty} x^2 \frac{(4x)^n}{n!}$$

$$\sum_{n=0}^{\infty} \frac{4^n x^{n+2}}{n!}$$

4. Find the Taylor series for $f(x) = \dfrac{1}{(3+x)^{3/2}}$ centered at $a = 1$

$f(x) = (3+x)^{-3/2}$

$f'(x) = -\frac{3}{2}(3+x)^{-5/2}$

$f''(x) = \frac{3\cdot 5}{2^2}(3+x)^{-7/2}$

$f'''(x) = \frac{-3\cdot 5\cdot 7}{2^3}(3+x)^{-9/2}$

$f(1) = 4^{-3/2} = \frac{1}{2^3}$

$f'(4) = -\frac{3}{2}\cdot\frac{1}{2^5}$

$f''(4) = \frac{3\cdot 5}{2^2}\,\frac{1}{2^7}$

$f'''(4) = \frac{-3\cdot 5\cdot 7}{2^3}\left(\frac{1}{2^9}\right)$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(1)\,(x-1)^n}{n!}$$

$$\underset{n=0}{\frac{1}{2^3}} - \underset{n=1}{\frac{1\cdot 3}{2^6}} + \underset{n=2}{\frac{1\cdot 3\cdot 5}{2^9}} - \underset{n=3}{\frac{1\cdot 3\cdot 5\cdot 7}{2^{12}}}$$

$$\frac{(-1)^n (1\cdot 3\cdot 5\cdots(2n+1))}{2^{3n+3}}$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (1\cdot 3\cdot 5\cdots(2n+1))(x-1)^n}{2^{3n+3}\, n!}$$

$$\lim_{n\to\infty} \left| \frac{\cancel{1\cdot 3\cdot 5\cdots(2n+1)}(2n+3)\cancel{(x-1)^n}}{2^3(\cancel{2^{3n+3}})(n+1)\,\cancel{n!}} \cdot \frac{\cancel{2^{3n+3}}\,\cancel{n!}}{\cancel{1\cdot 3\cdot 5\cdots(2n+1)}\,\cancel{(x-1)^n}} \right|$$

$$\lim_{n\to\infty} \left| \frac{2n+3}{8(n+1)}(x-1) \right| = \left| \frac{x-1}{4} \right| < 1$$

$|x-1| < 4$

$R = 4$

5. Find the sum of the power series a) $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{9^k (2k)!}$ $\sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{\pi}{3}\right)^{2k}}{(2k)!}$

$\cos(\pi/3) = 1/2$

b) $\sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{3^{2k}(2k+1)!}$ $\sum_{k=0}^{\infty} \frac{3}{\pi} \frac{(-1)^k \pi^{2k+1}}{3^{2k+1}(2k+1)!} =$

$\frac{3}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k (\pi/3)^{2k+1}}{(2k+1)!} = \frac{3}{\pi} \sin(\pi/3) = \frac{3}{\pi}\left(\frac{\sqrt{3}}{2}\right)$

6. Find the Taylor polynomial $T_4(x)$ for the function $f(x) = \sin x \ at \ a = 7\pi/6$ and use it to estimate $\sin(200\pi/180)$

$f(x) = \sin x$ $f(7\pi/6) = -1/2$

$f'(x) = \cos x$ $f'(7\pi/6) = -\sqrt{3}/2$

$f''(x) = -\sin x$ $f''(7\pi/6) = 1/2$

$f'''(x) = -\cos x$ $f'''(7\pi/6) = \sqrt{3}/2$

$T_3(x) = -\frac{1}{2} - \frac{\sqrt{3}}{2}(x - 7\pi/6) + \frac{1}{2}\frac{(x-7\pi/6)^2}{2!} + \frac{\sqrt{3}}{2}\frac{(x-7\pi/6)^3}{3!}$

$\frac{200\pi}{180} - \frac{7\pi}{6} = \frac{-10\pi}{180} = -\frac{\pi}{18}$

$\boxed{\sin(200\pi/180) \approx -\frac{1}{2} - \frac{\sqrt{3}}{2}\left(-\frac{\pi}{18}\right) + \frac{1}{2}\frac{(-\pi/18)^2}{2!} + \frac{\sqrt{3}}{2}\frac{(-\pi/18)^3}{3!} - \frac{1}{2}\frac{(-\pi/18)^4}{4!}}$

7. Estimate $\cos(0.1)$ to three non-zero terms using the know MacLaurin series.

$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$

$\cos(0.1) \approx 1 - \frac{(0.1)^2}{2!} + \frac{(0.1)^4}{4!}$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(1)(x-1)^n}{n!}$$

8. Use the **TAYLOR'S INEQUALITY** given below to

Approximate the function $f(x) = \dfrac{1}{(1+3x)^3}$ by a Taylor polynomial of degree 3 at $a = 1$

$f(x) = (1+3x)^{-3}$ | $f(1) = \frac{1}{4^3}$

$f'(x) = -3(1+3x)^{-4}(3)$ | $f'(1) = \frac{-3^2}{4^4}$

$f''(x) = 12(1+3x)^{-5}(9)$ | $f'' = \frac{12(9)}{4^5}$

$f'''(x) = -60(1+3x)^{-6}(27)$ | $f''' = \frac{-60(27)}{4^6}$

$$T_3(x) = \frac{1}{4^3} - \frac{9}{4^4}(x-1) + \frac{108}{4^5}\frac{(x-1)^2}{2!} - \frac{60(27)}{4^6}\frac{(x-1)^3}{3!}$$

How accurate is this approximation when $0.7 < x < 1.3$

$-.3 < x-1 < .3$

$|x-1| < .3$

$d = .3$

If $\left|f^{(n+1)}(x)\right| \le M$ for $|x-a| \le d$, then the remainder

$$R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1} \quad \text{for } |x-a| \le d$$

$f''''(x) = 360(1+3x)^{-7}(81)$

largest value $\dfrac{360(81)}{(1+3x)^7}$ on $0.7 < x < 1.3$

$$M \ge \left|\frac{360(81)}{(1+2.1)^7}\right| \text{ or } \left|\frac{29160}{(3.1)^7}\right|$$

$$R_3 \le \frac{\left|\frac{29,160}{(3.1)^7}\right| |.3|^4}{4!} = \frac{29,160}{2751.3}\frac{(.0081)}{24}$$